

define the symplectic rotation matrix

$$X(\theta_i) : \begin{bmatrix} x_i \\ p_i \end{bmatrix} \mapsto \begin{bmatrix} \cos(\theta_i) & \sin(\theta_i) \\ -\sin(\theta_i) & \cos(\theta_i) \end{bmatrix} \begin{bmatrix} x_i \\ p_i \end{bmatrix}. \quad (13)$$

Replacing B with $X(\theta_i)B$ in Eq. (10) leaves the trace invariant so there is a $SO(2)^n$ family of linear maps taking f_0 to a minimal energy configuration. This family of matrices is enlarged when the symplectic eigenvalues of either V or H are degenerate.

It is interesting to compare Eq. (8) and Equation Eq. (12). Notably, the AM-GM inequality shows that $E_{Sp(2n)} \geq E_{SL(2n)}$ since

$$2 \sum_{i=1}^n \lambda_i^H \lambda_{n+1-i}^V \geq 2n (\det(HV))^{1/2n}, \quad (14)$$

with equality iff $\lambda_i^H \lambda_{n+1-i}^V \equiv \text{const.}$ In particular, $E_{Sp(2)} = E_{SL(2)}$ as expected.

C. Example 1

We first compute an easy example in $n = 2$. For $\epsilon > 0$, suppose that $\mathcal{E}(\mathbf{z}, \epsilon) = x^2 + \epsilon^2 y^2 + p_x^2 + p_y^2$. Suppose further that $f_0(\mathbf{z}, R) = \frac{6}{R^2|B(R)|} \chi_{B(R)} = \frac{6}{R^2|B(R)|} \Theta(R^2 - |\mathbf{z}|^2)$ is a rescaled indicator function on the ball. We compute that $N = 1$, $\mathbf{c} = 0$, $H(R) = \mathbb{I}_4$, $V_0 = 0$, $\mathbf{d} = 0$, and $V(\epsilon) = \text{diag}(1, \epsilon^2, 1, 1)$. The initial energy stored in f_0 is $E[f_0] = (3 + \epsilon)$. For ϵ small, we should expect that $E_{SL(4)}$ is small since f_0 can be squeezed onto the y axis via area-preserving maps. Indeed, Eq. (8) gives us that

$$E_{SL(4)} = \epsilon^{1/2}. \quad (15)$$

Dividing this equation by $E[f_0]$, we can alternatively compute the inaccessible energy fraction $F_{SL(2n)} := \frac{E_{SL(2n)}}{E[f_0]}$ to be

$$F_{SL(4)} = \frac{4\epsilon^{1/2}}{3 + \epsilon^2}. \quad (16)$$

The distribution function after an energy minimizing linear map is $f_0 \circ \phi^{-1} = \frac{1}{|B(R)|} \Theta(R^2 - \epsilon^{-1/2}(x^2 + p_x^2 + p_y^2) - \epsilon^{3/2}y^2)$ which looks as expected.

In contrast, the linear Gromov's nonsqueezing theorem prohibits squeezing f_0 onto the y -axis via linear symplectomorphisms. This implies we should find $E_{Sp(2n)}$ to be finite in the limit $\epsilon \rightarrow 0^+$. $V(\epsilon)$ can be symplectically diagonalized by $S_V = \text{diag}(1, \epsilon^{-1/2}, 1, \epsilon^{1/2})$ giving